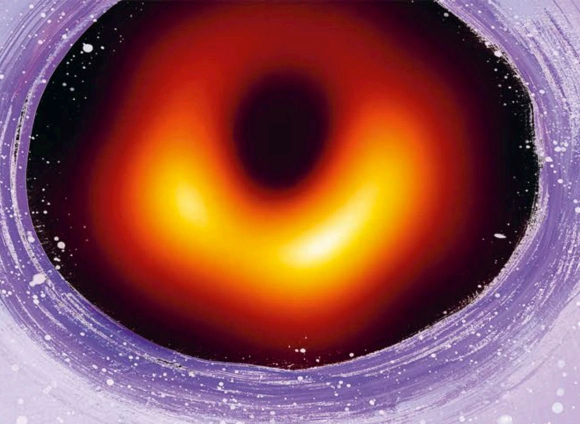
Hawking radiation

Stephen combined Albert Einstein's ideas about gravity with quantum theory to understand black holes better. His calculations showed that some radiation *can* escape from a black hole, so over time the old star will evaporate and disappear!



This is the first-ever photograph of a black hole. It was taken in 2019.

Black holes and beyond

As a massive star runs out of fuel, the nuclear reactions deep inside slow down. There is nothing to balance the star's huge gravitational pull, so the star collapses and all of its mass gets packed into a tiny lump. It was thought that the gravity around this lump is so incredibly strong that nothing can escape it—not even light! However, Stephen disproved this when he showed that *some* radiation can escape a black hole. Stephen visited black holes in his mind, using math to figure out what we may find there. He became known for his incredible scientific brain, but also for his talent at communicating science through books and talks.

Black holes are areas of space with such strong gravity that nothing seems to escape—not even light.

Black holes form when massive stars run out of fuel.

When motor neurone disease made it impossible for Stephen to speak, he began using a voice synthesizer. His electronic voice became famous around the world.